

MSO Undecidability for Hereditary Classes of Unbounded Clique-Width

Anuj Dawar

*Department of Computer Science and Technology,
University of Cambridge, U.K.*

Abhisekh Sankaran

Tata Consultancy Services Ltd., Mumbai, India

Abstract

Seese’s conjecture for finite graphs states that monadic second-order logic (MSO) is undecidable on all graph classes of unbounded clique-width. We show that to establish this it would suffice to show that grids of unbounded size can be interpreted in two families of graph classes: minimal hereditary classes of unbounded clique-width; and antichains of unbounded clique-width under the induced subgraph relation. We explore all the currently known classes of the former category and establish that grids of unbounded size can indeed be interpreted in them.

Keywords: clique width, Seese’s conjecture, antichain, MSO interpretation, grid

1. Introduction

The monadic second-order logic (MSO) of graphs has been an object of intensive research for many years now. It is a logic that is highly expressive and yet very well behaved on many interesting classes of graphs. It has enabled the
5 extension of many automata-theoretic and algebraic techniques to the construction of algorithms on graphs (see the comprehensive treatment in [1]). It has become a reference logic against which many others are compared. A key area of investigation is determining on which classes of graphs MSO is algorithmically well-behaved.

10 The good algorithmic behaviour of MSO on a class $\mathcal{C}$ of graphs is usually
 taken to mean one of two things: the evaluation (or model-checking) problem for
 MSO sentences on $\mathcal{C}$ is tractable; or the satisfiability problem of MSO sentences
 on $\mathcal{C}$ is decidable. Usually, these two are linked. Broadly speaking, the only
 way we know to show that the MSO theory of a class $\mathcal{C}$ is decidable is to show
 15 that $\mathcal{C}$ can be obtained by means of an MSO interpretation from a class of
 trees, which itself has a decidable theory and this also yields efficient evaluation
 algorithms for MSO sentences on $\mathcal{C}$. And the only way we know to show that the
 MSO theory of $\mathcal{C}$ is undecidable is to show that there is an MSO interpretation
 that yields arbitrarily large grids on $\mathcal{C}$ and, often this also yields an obstacle to
 20 the tractability of MSO evaluation on $\mathcal{C}$. There are exceptions to the latter in
 pathological cases (for instance, if the interpreted grid is much smaller than the
 structure in which it is interpreted) but [2] provides a fairly general instance of
 the rule.

Seese [3] formalizes the first of these observations into a conjecture: if the
 25 MSO theory of a class $\mathcal{C}$ is decidable, there is an MSO interpretation Ψ and
 a class $\mathcal{T}$ of trees such that Ψ maps $\mathcal{T}$ to $\mathcal{C}$. This remains an open question
 nearly three decades after it was first posed despite considerable research effort
 around it. By a theorem of Courcelle and Engelfriet [1], it is known that the
 classes of graphs obtained by MSO interpretations from trees are exactly those
 30 of bounded clique-width. Thus, Seese's conjecture can be understood as saying
 that any class of graphs of unbounded clique-width has an undecidable MSO
 theory. If we similarly formalize the second observation above about grids and
 combine it with this, we can formulate the following stronger conjecture, which
 we refer to below as the *strong Seese conjecture*: every class $\mathcal{C}$ of graphs of
 35 unbounded clique-width admits an MSO interpretation that defines arbitrarily
 large grids. Seese's conjecture is often formulated in this stronger form as it
 seems the only reasonable route to proving it. It can be seen as an interesting
 analogue of the Robertson-Seymour grid minor theorem to the effect that any
 class of graphs of unbounded treewidth admits arbitrarily large grids as minors.

40 In recent years there has been growing interest in clique-width as a measure

of the complexity of graphs from a structural and algorithmic point of view, quite separate from questions of logic [4, 5, 6, 7]. In particular, it provides a route for extending algorithmic methods that have had great success on sparse graph classes [8] to more general classes of graphs. A class of graphs may be of
45 bounded clique-width while containing dense graphs—the classic example being the class of cliques.

In the context of the structural study of classes of bounded clique-width, there is particular interest in *hereditary classes*, that is, classes of graphs closed under the operation of taking induced subgraphs. This is because the induced
50 subgraph relation behaves well with respect to clique-width. If a graph H is a subgraph or a minor of a graph G , the clique-width of H can be greater than that of G but if H is an *induced subgraph* of G , then the clique-width of H is no more than that of G . Hence, the hereditary closure of a class $\mathcal{C}$ of bounded clique-width still has bounded clique-width.

55 The induced subgraph relation is not as well-behaved as the graph minor relation. By the Robertson-Seymour graph minor theorem [9], the graph minor relation is a well-quasi-order. This is not true for the induced subgraph relation. By the same token, the classes of graphs of unbounded treewidth are well understood in that they are precisely the classes which have grid minors of
60 unbounded size. The picture for classes of graphs of unbounded clique-width is somewhat less clear. In particular, the relationship between a class having unbounded clique-width and admitting a well-quasi-order of the induced subgraph relation has been the subject of much investigation. It is possible to construct as we see below, infinite descending chains, under inclusion, of hereditary classes
65 of graphs, each of unbounded clique-width.

Lozin [10] identified the first example of a hereditary class $\mathcal{C}$ of graphs of unbounded clique-width that are *minimal* with this property—that is, no hereditary proper subclass of $\mathcal{C}$ has unbounded clique-width. Since then, many other such classes have been constructed. Collins et al. [11] show how to obtain
70 an infinite family of such classes. Their construction has been recently extended by Brignall and Cocks [12] to obtain uncountably many examples. Atminas et

al. [13] construct instances of such classes which are characterized by a finite collection of forbidden induced subgraphs. Lozin et al. [14] construct a minimal hereditary class of unbounded clique-width that is well-quasi-ordered under the induced subgraph relation.

This exploration of novel classes of unbounded clique-width also suggests an approach to establishing Seese’s conjecture for finite graphs. We establish in Section 3 that a proof of Seese’s conjecture would follow from the conjunction of the following two statements: (1) every collection of graphs of unbounded clique-width that forms an infinite anti-chain under the induced subgraph relation interprets arbitrarily large grids; and (2) every minimal hereditary class of unbounded clique-width interprets arbitrarily large grids. This suggests a programme to establish Seese’s conjecture by systematically studying antichains and minimal hereditary classes of unbounded clique-width. We do not yet know of a complete classification of minimal hereditary classes of unbounded clique-width, which makes a systematic approach to this programme challenging. Nevertheless, we examine in Sections 4–5 all known classes satisfying these conditions and show that in all cases we can indeed interpret grids of unbounded size. Thus none of these provides a counterexample to Seese’s conjecture. Our construction shows a uniform method of proving that these classes have unbounded clique-width. The proof is often simpler than the *ad hoc* methods by which this was proved for each class in the literature.

It is worth mentioning some significant lines of investigation related to Seese’s conjecture. Courcelle [15] shows that proving Seese’s conjecture for finite graphs is equivalent to proving the relativized version of the conjecture for particular classes of graphs, two examples being bipartite graphs and split graphs. He further shows the conjecture to be true when relativized to uniformly k -sparse graphs and interval graphs. Another line of work addresses variants of Seese’s conjecture obtained by considering logics other than MSO. One such result by Seese [3] shows that guarded second-order logic (GSO) is undecidable on any class of unbounded clique-width. Similarly, Courcelle and Oum [16] show that the extension $C_2\text{MSO}$ of MSO obtained by considering modulo 2 counting

quantifiers is also undecidable on classes of unbounded clique-width. In all of these cases, the proof goes via interpreting grids in unbounded clique-width
105 classes. There has also been interesting progress looking at Seese’s conjecture for structures other than graphs. A significant paper here is by Hliněný and Seese [17] who show the conjecture to be true for matroids representable over any finite field.

2. Preliminaries

110 The graphs we consider in this paper are simple, undirected and loop-free. For a graph G , we write $V(G)$ for the vertices of G and $E(G)$ for the edges. A graph H is an *induced subgraph* of G if $V(H) \subseteq V(G)$ and for any $x, y \in V(H)$, $\{x, y\} \in E(H)$ if, and only if, $\{x, y\} \in E(G)$. We write $H \subseteq G$ to denote that H is an induced subgraph of G . A class of graphs is said to be *hereditary* if
115 it is closed under induced subgraphs. For any graph class $\mathcal{C}$, we write $\mathcal{C} \downarrow$ to denote the hereditary closure of $\mathcal{C}$ – that is, the class of graphs that are induced subgraphs of the graphs in $\mathcal{C}$. We consider monadic second-order logic (MSO) over vocabularies τ containing the binary relation E and finitely many unary relation symbols. A τ -structure can be thought of as an *expansion* of a graph
120 $G = (V, E)$ with unary relations that interpret the unary symbols in τ . Such a structure can be thought of as a vertex-coloured graph. An MSO formula over the vocabulary τ is an expression that is inductively constructed from atomic MSO formulae using the Boolean connectives $\wedge, \vee$, and $\neg$, and existential quantification over vertex variables and set variables. Here an atomic MSO
125 formula is an expression of the form $E(x, y)$ or $Q(x)$ or $X(y)$ or $x = y$ where x, y are vertex variables, the predicates E, Q belong to τ and X is a set variable. A *first order*, or FO, formula is an MSO formula that does not contain any set variable. We often write $\varphi(\bar{x}, \bar{X})$ to denote a formula whose free variables are among $\bar{x}$ and $\bar{X}$, the former being a tuple of vertex variables and the latter a
130 tuple of set variables. Given such a formula, and a graph G along with a tuple $\bar{a}$ of vertices that interprets $\bar{x}$ and a tuple $\bar{A}$ of unary relations that interprets $\bar{X}$,

we write $(G, \bar{A}) \models \phi[\bar{a}]$ to denote that the formula ϕ is satisfied in G in this interpretation.

Given a graph G and an MSO formula $\varphi(\bar{x}, \bar{X})$ where the length of $\bar{x}$ is k ,
135 we can think of φ as defining a k -ary relation on an expansion of G with an interpretation $\bar{A}$ of $\bar{X}$. Specifically this relation, denoted $\varphi^{(G, \bar{A})}$, is given by $\varphi^{(G, \bar{A})} = \{\bar{a} \mid (G, \bar{A}) \models \varphi[\bar{a}]\}$. Given a sequence $\bar{Z}$ of set variables, an MSO *graph interpretation with parameters* $\bar{Z}$ is a pair $\Psi(\bar{Z}) = (\psi_V(x, \bar{Z}), \psi_E(x, y, \bar{Z}))$ of MSO formulas over the vocabulary $\{E\} \cup \{Z_i \mid Z_i \text{ is an element of } \bar{Z}\}$. Given
140 a graph G together with unary relations $\bar{A}$ interpreting the set variables $\bar{Z}$ in G , the interpretation $\Psi(\bar{Z})$ defines a possibly directed graph $H = \Psi(G, \bar{A})$. This graph has (i) vertex set $\psi_V^{(G, \bar{A})}$, and (ii) edge set $\psi_E^{(G, \bar{A})}$. In this paper, we are only interested in the case where $\Psi(\bar{Z})$ defines an undirected graph (that is, $\psi_E(x, y, \bar{Z})$ defines an irreflexive and symmetric binary relation). Thus $\Psi(\bar{Z})$
145 defines a function from the expansions of graphs with $|\bar{Z}|$ unary predicates, to graphs, and therefore in general defines a relation on graphs. Where it causes no confusion, we also refer to the relation defined by an interpretation as an MSO interpretation. If $\bar{Z}$ is empty, we call the interpretation Ψ *parameterless*, and such a Ψ defines a function from graphs to graphs. An example of a parameterless interpretation is $\Theta = (\theta_V(x), \theta_E(x, y))$ where $\theta_V(x) := (x = x)$ and $\theta_E(x, y) := \neg E(x, y)$; the function it defines maps a graph to its complement. An example of an interpretation with parameters is $\Gamma(Z) = (\gamma_V(x, Z), \gamma_E(x, y, Z))$ where $\gamma_V(x, Z) := Z(x)$ and $\gamma_E(x, y, Z) := E(x, y)$. The function that it defines maps an expansion (G, A) of a graph G to the subgraph of G induced by A ;
155 thus the relation on graphs that $\Gamma(Z)$ defines maps a graph to all its induced subgraphs. Given a class $\mathcal{C}$ of graphs and an interpretation Ψ with parameters $\bar{Z}$, we denote by $\Psi(\mathcal{C})$ the class of graphs given by $\Psi(\mathcal{C}) = \{\Psi(G, \bar{A}) \mid G \in \mathcal{C} \text{ and } \bar{A} \text{ is an interpretation of } \bar{Z} \text{ in } G\}$. For example, for the interpretation Γ above and a class $\mathcal{C}$ of graphs, the class $\Gamma(\mathcal{C})$ is exactly the hereditary closure of $\mathcal{C}$. Since they are relations, one can compose interpretations and it is known that the class of MSO interpretations is closed under composition [18].
160 We call MSO interpretations with parameters simply MSO interpretations for

ease of readability, and denote them with the uppercase Greek letters $\Phi, \Gamma, \Psi, \Theta$, etc.

165 The notion of clique-width is a structural parameter of graphs that was introduced by Courcelle, Engelfriet and Rozenberg in [19] as a generalization of the well-known notion of treewidth. We do not give the definitions of clique-width and treewidth here as we need only specific properties of these for our results that we state below; we point the reader to [1, 8] for more about the
 170 notions and results concerning them. We write $\text{cwd}(G)$ and $\text{twd}(G)$ for the clique-width and tree-width of a graph G , respectively. As examples, a clique has clique-width 1, and a cograph has clique-width 2. It is known for any graph G that $\text{cwd}(G) \leq 4 \cdot 2^{\text{twd}(G)-1} + 1$ [4] and for planar G we even have $\text{cwd}(G) \leq 6\text{twd}(G) - 2$ [20]. A class of graphs is said to have *bounded* clique-
 175 width if for some number $k \geq 1$, every graph in the class has clique-width at most k . Thus, the class of cliques, the class of cographs and all classes of bounded treewidth have bounded clique-width. A graph class has *unbounded* clique-width if it does not have bounded clique-width. Examples of graph classes of unbounded clique-width include grids, interval graphs, and line graphs [15].

180 The class of all graphs of clique-width at most k is hereditary since the clique-width of an induced subgraph of G is never more than the clique-width of G . An *antichain* under the induced subgraph relation is a set $\mathcal{A}$ of graphs such that if G and H are distinct graphs in $\mathcal{A}$, then neither of $G \subseteq H$ or $H \subseteq G$ holds. Usually when we say “antichain” without further qualification, we mean
 185 an antichain under the induced subgraph relation. A graph class $\mathcal{C}$ is said to be *well-quasi-ordered* (WQO) under induced subgraphs if it does not contain any infinite antichains. For example, the class of all cliques is WQO under induced subgraphs.

The MSO theory of a graph class $\mathcal{C}$ is the class of all MSO sentences that
 190 are true in all graphs of $\mathcal{C}$. This theory is decidable if, and only if, the following problem is decidable: given an MSO sentence ϕ decide if ϕ is true in some graph in $\mathcal{C}$. Seese’s conjecture states any class whose MSO theory is decidable has bounded clique-width. An $m \times n$ grid is a graph $G = (V, E)$ on $m \cdot n$ vertices

with $V = \{(i, j) \mid 1 \leq i \leq m, 1 \leq j \leq n\}$ and $E = \{\{(i, j), (i, j+1)\} \mid 1 \leq i \leq m, 1 \leq j < n\} \cup \{\{(i, j), (i+1, j)\} \mid 1 \leq i < m, 1 \leq j \leq n\}$. The grid is *square* if $m = n$. We
195 say a class $\mathcal{C}$ of graphs *interprets grids* via an MSO interpretation Φ , if $\Phi(\mathcal{C})$ contains graphs isomorphic to arbitrarily large square grids. Any class of graphs that contains arbitrarily large grids has undecidable MSO theory [1, Thm. 5.6]. Moreover, since MSO decidability is preserved by interpretations [1, Thm. 7.54],
200 any class of graphs that interprets grids via an MSO interpretation has an undecidable MSO theory. The *strong Seese conjecture* is that any class of unbounded clique-width interprets grids via an MSO interpretation. It is known that if the clique-width of a class $\mathcal{C}$ is bounded and Φ is an MSO interpretation, then the clique-width of $\Phi(\mathcal{C})$ is also bounded [1, Cor. 7.38]. A simple observation about
205 classes interpreting grids is the following.

Proposition 2.1. *Suppose $\mathcal{C}$ is a graph class that interprets grids, and $\mathcal{D}$ is a graph class for which there exists an MSO interpretation Ξ such that the hereditary closure of $\Xi(\mathcal{D})$ contains $\mathcal{C}$. Then $\mathcal{D}$ interprets grids as well.*

Specifically, if Θ is the interpretation mapping $\mathcal{C}$ to a class containing ar-
210 bitrarily large grids, and Γ is the interpretation defined above taking any class to its hereditary closure, then an interpretation Ω such that $\Omega(\mathcal{D})$ contains arbitrarily large square grids, is given by $\Omega = \Theta \circ \Gamma \circ \Xi$ (viewing Θ, Γ and Ξ as functions) where $\circ$ denotes composition.

We say that a class of graphs $\mathcal{C}$ is HUCW if it is hereditary and has un-
215 bounded clique-width. An HUCW graph class is said to be *minimal* if it does not contain a proper subclass that is HUCW. For example, bipartite permutation graphs and unit interval graphs are two minimal HUCW graph classes [10]. The existence of countably many minimal HUCW classes is established in [11], and this has been recently extended to uncountably many minimal HUCW classes
220 in [12].

3. Minimal Classes and Well-Quasi-Ordering

In this section we lay out an approach to studying Seese's conjecture that motivates our study of MSO decidability for minimal HUCW classes. The first observation is that, if $\mathcal{C}$ is a counter-example to Seese's conjecture, then so
 225 is $\mathcal{C} \downarrow$. Recall that a counter-example to Seese's conjecture would be a class $\mathcal{C}$ that has unbounded clique-width and a decidable MSO theory. Clearly if $\mathcal{C}$ has unbounded clique-width, then so does $\mathcal{C} \downarrow$. The following proposition is folklore. It follows immediately from the fact that MSO decidability is preserved by interpretations and the existence of the interpretation Γ defined above which
 230 takes a class to its hereditary closure.

Proposition 3.1. *If the MSO theory of $\mathcal{C}$ is decidable, then so is the MSO theory of $\mathcal{C} \downarrow$.*

Hence, if there is a counter-example to Seese's conjecture, we have one that is a hereditary class of unbounded clique-width, i.e. an HUCW class. In the present
 235 section, we establish some basic facts about the HUCW classes that allow us to structure the search for such a counter-example, or indeed the attempt to show that there is none.

The relation of being an induced subgraph is not a well-quasi-order as it admits infinite anti-chains. As an example, let I_n be the graph on $n+4$ vertices
 240 $e_0, e_1, e_2, e_3, c_1, \dots, c_n$ where for each $i < n$ there is an edge between c_i and c_{i+1} , and in addition we have edges $e_0 - c_1$, $e_1 - c_1$, $e_2 - c_n$ and $e_3 - c_n$. In short, there is a path of length n with two additional vertices at each end to mark the ends. Then, it is clear the collection $(I_n)_{n \in \mathbb{N}}$ is an antichain in the induced subgraph order. This particular antichain has bounded clique-width. It is also possible
 245 to construct antichains of unbounded clique-width (which therefore must be infinite). An example is obtained by taking the collection of $n \times n$ grids and adding two extra vertices at each corner to form a triangle. In what follows, whenever we refer to an *antichain* we mean one under the induced subgraph relation.

250 From an antichain of unbounded clique-width, it is possible to construct
(as we show below) an infinite descending chain of classes of graphs (under the
inclusion relation) all of which are HUCW. Thus, it was a significant discovery
to find that there are actually HUCW classes $\mathcal{C}$ that are *minimal*: no proper
hereditary subclass of $\mathcal{C}$ has unbounded clique-width. The first such example is
255 due to Lozin [10]. Collins et al. [11] constructed an infinite family of such classes
and Lozin et al. [14] give an example that is itself well-quasi-ordered under the
induced substructure relation. We examine these in some detail in subsequent
sections.

If it were the case that every class that is HUCW contains as a subclass a
260 minimal HUCW class, then showing that every minimal HUCW class interprets
grids would suffice to prove Seese’s conjecture. Indeed, if $\mathcal{C}$ interprets grids
of unbounded size, so does every class that contains $\mathcal{C}$. However, Korpelainen
has shown [21] that there are HUCW classes that contain no minimal HUCW
subclass. We give a construction of such a class in Section 3.2. This is linked to
265 the existence of antichains of unbounded clique-width. Specifically, we establish
the following facts.

1. If $\mathcal{C}$ is a minimal HUCW class, then it cannot contain an antichain of
unbounded clique-width (Theorem 3.5 in Section 3.1).
2. If $\mathcal{C}$ is an HUCW class that contains no minimal class, it must contain an
270 antichain of unbounded clique-width (Theorem 3.6 in Section 3.1).

From these, the theorem below follows, which suggests a programme for
proving Seese’s conjecture.

Theorem 3.2. *The strong Seese conjecture holds if, and only if, both of the
following are true:*

- 275 1. *every antichain of unbounded clique-width interprets grids; and*
2. *every minimal HUCW class interprets grids.*

3.1. Antichains and Minimal Classes

We first establish the relationship between the existence of antichains of unbounded clique-width and the minimality of HUCW classes. These are established in Theorems 3.5 and 3.6.

We say that a sequence $(\mathcal{C}_i)_{i \in \omega}$ is an infinite *strictly descending* HUCW-chain if for each i , $\mathcal{C}_i$ is an HUCW class and $\mathcal{C}_{i+1}$ is a proper subclass of $\mathcal{C}_i$. We say that $\mathcal{C}$ contains an infinite strictly descending HUCW-chain if there is such a chain with $\mathcal{C}_i \subseteq \mathcal{C}$ for all i .

Lemma 3.3. *The following are equivalent:*

1. $\mathcal{C}$ contains an infinite strictly descending HUCW-chain whose intersection is a class of bounded clique-width.
2. $\mathcal{C}$ contains an infinite strictly descending HUCW-chain whose intersection is empty.
3. $\mathcal{C}$ contains an antichain of unbounded clique-width.

Proof. (3) $\rightarrow$ (2): If $\{G_1, G_2, \dots\}$ is such an antichain, then let $\mathcal{C}_i$ be the hereditary closure of $\{G_i, G_{i+1}, \dots\}$ for $i \geq 1$. Then $\mathcal{C}_1 \supsetneq \mathcal{C}_2 \supsetneq \dots$ is an infinite strictly descending HUCW-chain whose intersection is empty.

(2) $\rightarrow$ (1): Trivial since the empty class has clique-width 0.

(1) $\rightarrow$ (3): Let $\mathcal{C}_1 \supsetneq \mathcal{C}_2 \supsetneq \dots$ be such a descending HUCW-chain and $\mathcal{C}_\omega = \bigcap_{i \geq 1} \mathcal{C}_i$. Let $\mathcal{D}_i = \mathcal{C}_i \setminus \mathcal{C}_{i+1}$ for $i \geq 1$. Then for $1 \leq i < j$, we have $\mathcal{D}_i \cap \mathcal{C}_j = \emptyset$; hence $\mathcal{D}_i \cap \mathcal{D}_j = \mathcal{D}_i \cap \mathcal{C}_\omega = \emptyset$. Further, $\mathcal{C}_i = \left(\biguplus_{1 \leq k < \omega} \mathcal{D}_k \right) \uplus \mathcal{C}_\omega$.

Claim 3.4. *The following are true:*

1. For $1 \leq i < j$, no graph in $\mathcal{D}_i$ is an induced subgraph of a graph in $\mathcal{D}_j$.
2. For $i \geq 1$, for every graph $G \in \mathcal{D}_i$, there exists a number $f(G) > i$ such that for all $j \geq f(G)$, no graph in $\mathcal{C}_j \setminus \mathcal{C}_\omega$ is an induced subgraph of G .

Proof. (1): If $G \subseteq H$ for some $G \in \mathcal{D}_i$ and $H \in \mathcal{D}_j$, then since $\mathcal{D}_j \subseteq \mathcal{C}_j$ and $\mathcal{C}_j$ is hereditary, we would have $G \in \mathcal{C}_j$; but that contradicts the fact that $\mathcal{D}_i \cap \mathcal{C}_j = \emptyset$.

(2): Let $H_1, \dots, H_r$ be an enumeration of the induced subgraphs of G that
 305 are not in $\mathcal{C}_\omega$ – clearly r is finite since G is finite. Since $\mathcal{C}_i = (\uplus_{i \leq j < \omega} \mathcal{D}_j) \uplus \mathcal{C}_\omega$,
 there exist numbers $j_1, \dots, j_r \in [i, \omega)$ such that $H_i \in \mathcal{D}_{j_i}$ for $i \in \{1, \dots, r\}$. It then
 follows by the properties of the $\mathcal{D}_i$'s above that $f(G) = \max\{j_i \mid 1 \leq i \leq r\} + 1$ is
 indeed as desired. $\square$

We now use the above claim to inductively construct an antichain of $\mathcal{C}$
 310 of unbounded clique-width. Let G_0 be a graph in $\mathcal{D}_0$. Assume that we have
 constructed graphs $G_0, \dots, G_i$ for $i \geq 0$ such that (i) $G_j \in \mathcal{D}_{l_j}$ and $l_j > l_{j-1}$ for
 $1 \leq j \leq i$; (ii) $\{G_0, \dots, G_i\}$ is an antichain; and (iii) the clique-width of G_j is
 strictly greater than that of G_{j-1} for $1 \leq j \leq i$. Let $k = \max\{f(G_j) \mid 1 \leq j \leq i\} > l_i$
 where f is as in Claim 3.4. Consider the class $\mathcal{C}_k \setminus \mathcal{C}_\omega$ – by Lemma 3.4, all graphs
 315 in this class are incomparable with each of $G_0, \dots, G_i$ in the induced subgraph
 order. Further, since $\mathcal{C}_k$ has unbounded clique-width while $\mathcal{C}_\omega$ has bounded
 clique width, we have that $\mathcal{C}_k \setminus \mathcal{C}_\omega$ has unbounded clique-width, whereby there
 exists $G_{i+1} \in \mathcal{C}_k \setminus \mathcal{C}_\omega$ such that G_{i+1} has clique-width greater than that of G_i .
 Let $l_{i+1} \geq k > l_i$ be such that $G_{i+1} \in \mathcal{D}_{l_{i+1}}$. Then we see that G_{i+1} is indeed as
 320 desired to complete the induction. $\square$

We are now ready to prove the two results linking minimality of HUCW
 classes and the existence of antichains of unbounded clique-width.

Theorem 3.5. *If $\mathcal{C}$ is a minimal HUCW class, then $\mathcal{C}$ does not contain an
 antichain of unbounded clique-width.*

325 *Proof.* If $\mathcal{C}$ contains an antichain of unbounded clique-width, then by Lemma 3.3,
 we have that $\mathcal{C}$ contains an infinite strictly descending HUCW-chain, and hence
 in particular a proper subclass that is HUCW. Hence $\mathcal{C}$ is not minimal. $\square$

Theorem 3.6. *If $\mathcal{C}$ is HUCW and does not contain a minimal HUCW class,
 then there exists in $\mathcal{C}$ an antichain of unbounded clique-width.*

330 *Proof.* We assume without loss of generality that the vertices of the graphs of $\mathcal{C}$
 belong to the set $\mathbb{N}$ of natural numbers, so that $\mathcal{C}$ is countable. Suppose that $\mathcal{C}$

does not contain a minimal class. Consider the sequence $(\mathcal{C}_\lambda)_{\lambda \geq 0}$ of classes of structures, for ordinals λ , defined inductively as follows. Let $\mathcal{C}_0 = \mathcal{C}$ and inductively, assume that for all $\nu < \lambda$, the class $\mathcal{C}_\nu$ has been defined and that
335 $\mathcal{C}_\nu \subseteq \mathcal{C}$ for all $\nu < \lambda$. If λ is a limit ordinal, define $\mathcal{C}_\lambda = \bigcap_{\nu < \lambda} \mathcal{C}_\nu$. If λ is a successor ordinal of say λ^- , then define $\mathcal{C}_\lambda$ as follows. If $\mathcal{C}_{\lambda^-}$ is not HUCW, then $\mathcal{C}_\lambda = \mathcal{C}_{\lambda^-}$. Otherwise $\mathcal{C}_{\lambda^-}$ is HUCW and $\mathcal{C}_{\lambda^-} \subseteq \mathcal{C}$; then $\mathcal{C}_{\lambda^-}$ cannot be minimal since by our premise, $\mathcal{C}$ does not contain any minimal HUCW class. Let $\mathcal{C}_\lambda$ be any proper subclass of $\mathcal{C}_{\lambda^-}$ that is HUCW. This completes the construction of
340 the sequence $(\mathcal{C}_\lambda)_{\lambda \geq 0}$.

Consider now the set $\mathcal{P}$ of ordinals defined as $\mathcal{P} = \{\lambda \mid \mathcal{C}_\lambda \text{ is not HUCW}\}$. This set is non-empty – since $\mathcal{C}$ is a class of finite graphs whose vertices are natural numbers, $\mathcal{C}$ is countable and hence $\mathcal{C}_\lambda = \emptyset$ for all uncountable λ . By the definition above, if $\lambda \in \mathcal{P}$, then all ordinals greater than λ are in $\mathcal{P}$ as well.
345 Now since the ordinals are well ordered, $\mathcal{P}$ has a minimum, call it μ^* . We make the following observations about μ^* :

1. μ^* must be a limit ordinal. If it is a successor ordinal of say λ , then $\mathcal{C}_\lambda$ must be HUCW since μ^* is the minimum ordinal in $\mathcal{P}$. But if $\mathcal{C}_\lambda$ is HUCW, then $\mathcal{C}_{\mu^*}$ must be a HUCW class by the inductive definitions above.
350 Therefore, $\mathcal{C}_{\mu^*} = \bigcap_{\nu < \mu^*} \mathcal{C}_\nu$ where $\mathcal{C}_\nu$ is HUCW for all $\nu < \mu^*$.
2. μ^* is countable – this is because $\mathcal{C}$ is countable.
3. $\mathcal{C}_{\mu^*}$ is a hereditary class of bounded clique-width. Let $G \in \mathcal{C}_{\mu^*}$ and $H \subseteq G$. Then by (1) above, $G \in \mathcal{C}_\nu$ for all $\nu < \mu^*$. Since each $\mathcal{C}_\nu$ is hereditary, we have $H \in \mathcal{C}_\nu$ for all $\nu < \mu^*$. Then $H \in \mathcal{C}_{\mu^*}$. So $\mathcal{C}_{\mu^*}$ is hereditary. That $\mathcal{C}_{\mu^*}$
355 has bounded clique-width now follows from the fact that $\mathcal{C}_{\mu^*}$ is not HUCW.

Now since μ^* is countable, it has cofinality ω so that there exists an increasing function $f : \mathbb{N} \rightarrow \mu^*$ (where μ^* is seen as the set of ordinals less than μ^*) such that if $\mathcal{F}_i = \mathcal{C}_{f(i)}$ for $i \in \mathbb{N}$, then $\bigcap_{i \in \mathbb{N}} \mathcal{F}_i = \mathcal{C}_{\mu^*}$. We observe that $\mathcal{F}_1 \not\supseteq \mathcal{F}_2 \not\supseteq \dots$ is an infinite strictly descending HUCW-chain in $\mathcal{C}$, whose intersection $\mathcal{C}_{\mu^*}$ is a
360 class of bounded clique-width. It now follows by Lemma 3.3 that $\mathcal{C}$ contains an

antichain of unbounded clique-width. $\square$

The converse of Theorem 3.6 does not hold. That is to say, we can construct an HUCW class that both contains a minimal HUCW class and contains an antichain of unbounded clique-width. Indeed, if $\mathcal{C}_1$ is a minimal HUCW class and $\mathcal{C}_2$ the hereditary closure of an antichain of unbounded clique-width then clearly $\mathcal{C} = \mathcal{C}_1 \cup \mathcal{C}_2$ has this property.

3.2. HUCW Classes which Contain No Minimal Class

Theorem 3.6 raises the obvious question of whether there exists any class $\mathcal{C}$ which is HUCW but does not contain a minimal HUCW class. The existence of such a class was demonstrated by Korpelainen [21]. Here we give a similar construction which we arrived at independently.

Theorem 3.7. *There is an HUCW class $\mathcal{T}$ that does not contain any minimal HUCW class.*

It suffices to show that if $\mathcal{C}$ is any hereditary subclass of $\mathcal{T}$ of unbounded clique-width, it contains an antichain of unbounded clique-width.

Towards this, let $G_{n,n}$ denote the $n \times n$ grid. Note that, in $G_{n,n}$, every vertex has degree 2, 3 or 4, and there are exactly four vertices (at the corners) of degree 2. For $n \geq 3$, we define T_n as the graph obtained from $G_{n,n}$ by:

1. removing every vertex v of degree 2 and inserting an edge between the two neighbours of v ; and
2. replacing every vertex v of degree 4 by four new vertices v_1, v_2, v_3, v_4 that are connected in a 4-cycle so that the four edges incident on v are now each incident on one of the four new vertices.

It is easily seen that T_n is 3-regular, and it is more convenient to work with than grids. The number of vertices in T_n is less than $4n^2$.

Recall that a graph H is a *subdivision* of a graph G if it is obtained from G by replacing every edge by a simple path. For a positive integer t , we write G^t

for the t -subdivision of G : the graph obtained from G by replacing each edge of G by a path of length t . We make the following simple observation for later

390 use:

Lemma 3.8. *If H is a subdivision of G and $\text{twd}(G) = k$, then $k \leq \text{twd}(H) \leq \max(k, 3)$.*

Proof. The lower bound on $\text{twd}(H)$ follows immediately from the fact that G is a minor of H so $\text{twd}(G) \leq \text{twd}(H)$.

395 Suppose now that (T, β) is a tree decomposition of G of width k . To obtain a tree decomposition of H , consider an edge $\{u, v\}$ of G which is subdivided into a path $u = p_0, \dots, p_t = v$ in H . As $\{u, v\}$ is an edge of G , there must be a node a of T such that $\{u, v\} \subseteq \beta(t)$. We attach a path $a_1, \dots, a_t$ of length t to a and let $\beta(a_i) = \{u, v, p_i, p_{i+1}\}$. Doing this for each edge gives us a tree decomposition
400 of H whose width is $\max(k, 3)$. $\square$

Define the class $\mathcal{T} = \{H \mid H \subseteq T_n^n \text{ for some } n > 2\}$, i.e. the collection of graphs that are induced subgraphs of the n -subdivision of T_n for some n . We consider the graphs $H \in \mathcal{T}$ where every vertex has degree 2 or 3. We call such graphs *skeleton graphs* and the vertices of degree 3 the *branch vertices*. Note
405 that every graph in $\mathcal{T}$ is an induced subgraph of a skeleton graph.

The next two lemmas establish some useful properties of the graphs in $\mathcal{T}$.

Lemma 3.9. *If $H \in \mathcal{T}$ is a skeleton graph with at most $m > 2$ branch vertices, then $\text{cwd}(H) \leq 6m - 2$.*

Proof. Since H has at most m branch vertices, it is the subdivision of some
410 graph G with m vertices. Hence, by Lemma 3.8, the treewidth of H is at most m . Note further that all graphs in $\mathcal{T}$ are planar and hence H is planar. For any planar graph H , $\text{cwd}(H) \leq 6\text{twd}(H) - 2$ [20, Thm 17], and the result follows. $\square$

Lemma 3.10. *If H is a subdivision of T_n for $n > 2$, then the clique-width of H
415 is at least $(n - 1)/6$.*

Proof. Since $G_{n-2,n-2}$ is a minor of T_n and $\text{twd}(G_{k,k}) = k$ we have that $\text{twd}(T_n) \geq n-2$. Also, by Lemma 3.8 we know that $\text{twd}(H) = \text{twd}(T_n)$. Now, for any planar graph G we have $\text{twd}(G) \leq 6\text{cwd}(G) - 1$ by [1, Prop. 2.115]. Since H is planar, the result follows. $\square$

420 *Proof of Theorem 3.7.* The class $\mathcal{T}$ is hereditary by definition and has unbounded clique-width by Lemma 3.10. Thus, it remains to show that for every class $\mathcal{C} \subseteq \mathcal{T}$, if $\mathcal{C}$ has unbounded clique-width, then $\mathcal{C}$ contains an antichain of unbounded clique-width.

So, suppose $\mathcal{C} \subseteq \mathcal{T}$ has unbounded clique-width. For a graph $H \in \mathcal{T}$, write
 425 $\text{mn}(H)$ for the length of the shortest path between two branch vertices of H . We define the following sequence of graphs. First, let G_0 be any graph in $\mathcal{C}$ containing at least two branch vertices. Suppose we have defined G_i for $i \geq 0$, and let $t = \max(\text{cwd}(G_i), \text{mn}(G_i))$. We then choose G_{i+1} to be any graph in $\mathcal{C}$ with $\text{cwd}(G_{i+1}) > 24t^2 - 2$.

430 It is clear that the sequence of graphs $(G_i : i \in \omega)$ is of unbounded clique-width, since $\text{cwd}(G_i) < \text{cwd}(G_{i+1})$ for all i . We now argue that this is also an antichain. For any $i < j$, clearly G_j cannot be an induced subgraph of G_i since $\text{cwd}(G_i) < \text{cwd}(G_j)$, so it remains to show that G_i is not an induced subgraph of G_j . Since $\text{cwd}(G_j) > 24t^2 - 2$, where $t = \max(\text{cwd}(G_i), \text{mn}(G_i))$, it follows
 435 by Lemma 3.9 that G_j has more than $4t^2$ branch vertices. Since T_n^n contains fewer than $4n^2$ branch vertices, it follows that G_j is not an induced subgraph of T_n^n for any $n \leq t$. Hence, $\text{mn}(G_j)$ is at least $t + 1$. However, by the choice of t , $\text{mn}(G_i) \leq t$ and so G_i contains two branch vertices at distance at most t . We conclude that G_i is not an induced subgraph of G_j . $\square$

440 4. Grid-Like Classes

We begin our systematic exploration of all known minimal hereditary classes of unbounded clique-width. Many such classes are defined in terms of a grid-like structure and this is used to show that they have unbounded clique-width. The challenge in these cases is to show how this grid structure can be drawn out

445 through an MSO interpretation. We begin with a collection of minimal HUCW classes (indeed, an uncountable collection of them) defined in terms of certain infinite words and show in Section 4.1 that they interpret grids. This is then extended by reductions in Section 4.2 to a number of other classes.

4.1. Word-defined minimal classes

450 Our starting point is a construction given by Brignall and Cocks [12] to demonstrate that there are uncountably many minimal HUCW classes, extending a construction by Collins et al. [11] showing the existence of infinitely many such classes. They construct a hereditary class $\mathcal{S}_\alpha$ of graphs for each ω -word $\alpha \in \{0, 1, 2, 3\}^\omega$ and show that as long as α contains infinitely many non-zero letters, the class $\mathcal{S}_\alpha$ has unbounded clique-width. 455 Moreover, for uncountably many distinct such α , $\mathcal{S}_\alpha$ is also minimal. The conditions under which $\mathcal{S}_\alpha$ is minimal need not concern us here. We are able to show that whenever α contains infinitely many non-zero letters $\mathcal{S}_\alpha$ interprets grids via MSO interpretations. In particular, this covers all minimal classes $\mathcal{S}_\alpha$ of unbounded clique-width, 460 including those defined in [11]. Before we proceed to a proof, we give a precise definition of the classes $\mathcal{S}_\alpha$.

The class $\mathcal{S}_\alpha$ is defined as the class of all finite induced subgraphs of a single countably infinite graph $\mathcal{P}_\alpha$. The set of vertices of $\mathcal{P}_\alpha$ is $\{v_{i,j} \mid i, j \in \mathbb{N}\}$. We think of the set as an infinite collection of *columns* $V_j = \{v_{i,j} \mid i \in \mathbb{N}\}$. All 465 edges are between vertices in adjacent columns, i.e. there is no edge between $v_{i,j}$ and $v_{i',j'}$ unless $j' = j + 1$ or $j' = j - 1$. The edges between successive columns are defined by the word α according to the following rules.

1. If $\alpha_j = 0$, then $\{v_{i,j}, v_{k,j+1}\} \in E(\mathcal{P}_\alpha)$ if, and only if, $i = k$.
2. If $\alpha_j = 1$, then $\{v_{i,j}, v_{k,j+1}\} \in E(\mathcal{P}_\alpha)$ if, and only if, $i \neq k$ for $i, k \in \mathbb{N}$.
- 470 3. If $\alpha_j = 2$, then $\{v_{i,j}, v_{k,j+1}\} \in E(\mathcal{P}_\alpha)$ if, and only if, $i \leq k$ for $i, k \in \mathbb{N}$.
4. If $\alpha_j = 3$, then $\{v_{i,j}, v_{k,j+1}\} \in E(\mathcal{P}_\alpha)$ if, and only if, $i \geq k$ for $i, k \in \mathbb{N}$.

The class $\mathcal{S}_\alpha$ is now given by $\mathcal{S}_\alpha = \{G \mid G \text{ is a finite induced subgraph of } \mathcal{P}_\alpha\}$.

We show the following theorem in this section.

Theorem 4.1. *Let $\alpha \in \{0, 1, 2, 3\}^\omega$ be such that α contains infinitely many non-zero letters. Then there exists an MSO interpretation Θ such that $\Theta(\mathcal{S}_\alpha)$ contains the class of all square grids.*

To prove Theorem 4.1, we show the existence of an MSO interpretation Ψ such that the hereditary closure of $\Psi(\mathcal{S}_\alpha)$ contains the class of all square grids. Proposition 2.1 ensures that this indeed suffices. It is clear that graphs in $\mathcal{S}_\alpha$ have a built-in grid-like structure with vertices arranged in rows and columns. The main challenge is to show that a sufficient part of this structure can be made explicit using an MSO interpretation. We give an outline of the construction.

What we show is that we can find in $\mathcal{S}_\alpha$ a sequence of graphs G_n for $n \in \mathbb{N}$ within which we can interpret *upper triangular grids*. One can think of an upper triangular grid U_t as the subgraph of the $t \times t$ grid induced by the vertices above the main diagonal, i.e. those vertices in the set $\{(i, j) \mid 1 \leq i, j \leq t\}$ with $i \leq j$. It is clear that U_t has as an induced subgraph an $r \times r$ grid, where $r = \lfloor \frac{t}{2} \rfloor$.

Let $\alpha \in \{0, 1, 2, 3\}^\omega$ be an ω -word containing infinitely many non-zero letters. We write α_i for the i^{th} letter of α . Let $p < \omega$ be the least value such that $\alpha_p \neq 0$. Fix $n \geq 1$ and let l be the length of the shortest contiguous subsequence of α starting at α_p that contains exactly $2n + 2$ elements which are not 0. We write $\beta_0 \cdots \beta_{l-1}$ for this sequence, so $\beta_0 = \alpha_p$.

Recall that the vertices of P_α are $\{v_{i,j} \mid i, j \in \mathbb{N}\}$, and we write V_j for the set $\{v_{i,j} \mid i \in \mathbb{N}\}$. We define the graph G_n to be the subgraph of P_α induced by the set $C = \bigcup_{i=0}^{l-1} C_i$ where $C_i \subseteq V_{p+i}$ is defined as follows for $0 \leq i < l$.

1. $C_0 = \{v_{0,p}, v_{1,p}, v_{2,p}, v_{3,p}\}$; and
2. $C_{i+1} = \{v_{0,p+i+1}, v_{1,p+i+1}, \dots, v_{t-1,p+i+1}\}$ where $t = |C_i|$ if $\beta_{i+1} = 0$ and $t = |C_i| + 1$ otherwise.

It is clear that $G_n \in \mathcal{S}_\alpha$. We show that we can interpret upper triangular grids in this class of graphs. The key challenge in defining the required inter-

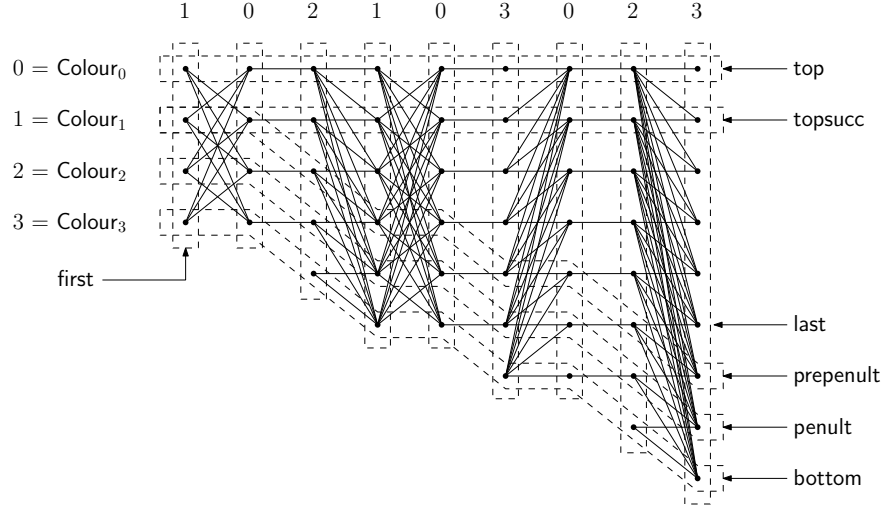


Figure 1: The graph H_2 for $\alpha = (102103023)^\omega$. The unlabeled graph underlying H_2 is G_2 .

pretation is to define the two binary relations: one that relates vertices that are in the same column and the other that relates vertices that are in the same row. In constructing the interpretation we make use of a number of set parameters to obtain a labeled version H_n of G_n ; in particular, H_n uses unary predicates for the vertices corresponding to the possible values of β_i , for the first and last column, the top, bottom and penultimate rows, and the rows immediately succeeding and preceding the top and penultimate rows respectively. The “diagonal” nature of the bottom row is vital to allowing us to define when two vertices are in successive columns, which we need in order to define the two relations of being in the same row and in the same column. We now give all the technical details below.

We define the graph H_n as the expansion of G_n with unary predicates $\text{Colour}_0, \text{Colour}_1, \text{Colour}_2, \text{Colour}_3, \text{top}, \text{topsucc}, \text{bottom}, \text{penult}, \text{prepenult}, \text{first}$ and last which are interpreted as follows. For $i \in \{0, 1, 2, 3\}$, the predicate Colour_i is interpreted as the set $\bigcup_{\beta(j)=i} C_j$; **top** is interpreted as the top row of G_n ; **topsucc** is interpreted as the second row of G_n after the top; **bottom** is the set of all vertices $v_{i,j}$ such that $i' \leq i$ for all $v_{i',j} \in C_j$; **penult** is the set of all vertices $v_{i,j}$ such

that $v_{i+1,j}$ is in **bottom**; **prepenult** is the set of all vertices $v_{i,j}$ such that $v_{i+2,j}$ is in **bottom**; and finally, **first** and **last** are interpreted as the sets C_0 and C_{l-1} respectively. Figure 1 provides an illustration.

We now describe the construction of the interpretation Ψ . Towards this, we need a number of auxiliary predicates which we define below.

1. **samecolumn**(x, y): This predicate is true of x, y in H_n if $x, y \in C_i$ for some $i \in \{0, \dots, l-2\}$ and $\beta_i \in \{1, 2, 3\}$ as long as neither of x, y is in **top** or **bottom**.

$$\begin{aligned}
\text{H-boundary}(x) &:= \text{top}(x) \vee \text{bottom}(x) \\
\text{samecolumn}(x, y) &:= \neg(\text{H-boundary}(x) \vee \text{H-boundary}(y)) \wedge \\
&\quad \neg \text{Colour}_0(x) \wedge \bigwedge_{i=1}^{i=3} \text{Colour}_i(x) \leftrightarrow \text{Colour}_i(y) \wedge \\
&\quad \forall z ((\neg \text{Colour}_3(x) \wedge \text{bottom}(z)) \vee \\
&\quad (\text{Colour}_3(x) \wedge \text{top}(z))) \\
&\quad \rightarrow (E(x, z) \leftrightarrow E(y, z))
\end{aligned}$$

To understand the last condition, note that if x and y are in the same column C_i that is not the last, with $\beta_i \in \{1, 2, 3\}$, and neither of x or y is in the top or bottom row in C_i , then the bottom elements of C_{i-1} and C_{i+1} are neighbours of either both x and y or neither; likewise for the top elements of C_{i-1} and C_{i+1} . On the other hand, suppose x and y are in different columns, say C_i and C_j respectively with $i < j$. Since β_j is 1, 2 or 3, we know that $|C_{j+1}| = |C_j| + 1$, and hence if β_j is 1 or 2, then every element of C_j , in particular y , is adjacent to the bottom element z of C_{j+1} , and if β_j is 3, then every element of C_j , and y in particular, is adjacent to the top element z' of C_{j+1} . Since x is not in a column adjacent to C_{j+1} , it cannot have an edge to either z or z' , and hence x and y do not satisfy the predicate **samecolumn**.

2. **adjcolumn**(x, y): This predicate is true of x, y in H_n if for some i, j with $|i-j| = 1$ and $\beta_i \in \{1, 2, 3\}$, it holds that x is in C_i but not in **top** or **bottom**,

and $y \in C_j$.

$$\begin{aligned} \text{adjcolumn}(x, y) &:= \neg(\text{H-boundary}(x) \vee \text{Colour}_0(x)) \wedge \\ &\quad \exists u (\text{samecolumn}(x, u) \wedge E(u, y)) \end{aligned}$$

3. $\text{domain}(x)$; This predicate is true of x in H_n if it is not one of the “periphery” vertices of G_n .

$$\text{domain}(x) := \neg(\text{H-boundary}(x) \vee \text{topsucc}(x) \vee \text{penult}(x) \vee \text{first}(x) \vee \text{last}(x))$$

4. $\text{rhscolumn}[i; S](x, y)$: For $i \in \{0, 1, 2, 3\}$ and $S \subseteq \{0, 1, 2, 3\}$, this predicate is true of x, y in H_n if $\text{domain}(x)$ and $\text{domain}(y)$ both hold and if $x \in C_j$ and $y \in C_{j+1}$ for some j with $\beta_j = i$ and $\beta_{j+1} \in S$. We need this predicate only for the following specific values of $[i; S]$: (i) $[0; \{1, 2, 3\}]$, (ii) $[1; \{0, 2, 3\}]$, (iii) $[2; \{0, 1, 2, 3\}]$, and (iv) $[3; \{0, 1, 2, 3\}]$.

$$\begin{aligned} \text{rhscolumn}[3; \{0, 1, 2\}](x, y) &:= \text{domain}(x) \wedge \text{Colour}_3(x) \wedge \text{domain}(y) \wedge \\ &\quad \bigwedge_{i=0}^{i=3} \text{Colour}_i(y) \rightarrow \eta_i(x, y) \\ \eta_0(x, y) &:= \text{adjcolumn}(x, y) \wedge \exists v (\text{samecolumn}(x, v) \\ &\quad \wedge \text{penult}(v) \wedge E(v, y)) \\ \eta_1(x, y) &:= \text{adjcolumn}(x, y) \wedge \\ &\quad \exists v (\text{samecolumn}(x, v) \wedge \text{prepenult}(v) \wedge \\ &\quad \exists w (\text{samecolumn}(y, w) \wedge \neg E(w, v))) \\ \eta_2(x, y) &:= \eta_1(x, y) \\ \eta_3(x, y) &:= \text{adjcolumn}(x, y) \wedge \\ &\quad \exists v (\text{samecolumn}(x, v) \wedge \text{prepenult}(v) \wedge \\ &\quad \exists w (\text{samecolumn}(y, w) \wedge E(w, v))) \end{aligned}$$

$$\begin{aligned}
\text{rhscolumn}[2; \{0, 1, 2\}](x, y) &:= \text{domain}(x) \wedge \text{Colour}_2(x) \wedge \text{domain}(y) \wedge \\
&\quad \bigwedge_{i=0}^{i=3} \text{Colour}_i(y) \rightarrow \eta_i(x, y) \\
\text{rhscolumn}[1; \{0, 2, 3\}](x, y) &:= \text{domain}(x) \wedge \text{Colour}_1(x) \wedge \text{domain}(y) \wedge \\
&\quad \neg \text{Colour}_1(y) \wedge \\
&\quad \bigwedge_{i \in \{0, 2, 3\}} \text{Colour}_i(y) \rightarrow \eta_i(x, y) \\
\text{rhscolumn}[0; \{1, 2, 3\}](x, y) &:= \text{domain}(x) \wedge \text{Colour}_0(x) \wedge \text{domain}(y) \wedge \\
&\quad \neg \text{Colour}_0(y) \wedge \text{adjcolumn}(y, x) \wedge \\
&\quad \forall v (\text{samecolumn}(y, v) \wedge E(v, x)) \rightarrow \\
&\quad \neg((\text{Colour}_3(v) \wedge \text{penult}(v)) \vee \\
&\quad (\text{Colour}_2(v) \wedge \text{topsucc}(v)))
\end{aligned}$$

These predicates are meant to give an orientation to some edges in the
550 symmetric relation $\text{adjcolumn}(x, y)$. Thus, it is sufficient to argue that if x is in C_i , then y cannot be in C_{i-1} . We present the argument for the case when $\beta_i = 3$. Other cases can be argued similarly.

Suppose $\beta_i = 3$. There are four subcases depending on the value of β_{i-1} . If $\beta_{i-1} = 0$, then the only element z of C_{i-1} that is adjacent to the penultimate
555 element of C_i is the bottom element of C_{i-1} . But then $\text{domain}(z)$ does not hold. Thus no $y \in C_{i-1}$ satisfies the formula $\eta_0(x, y)$. If $\beta_{i-1} \in \{1, 2\}$, then the only element of C_{i-1} that is not adjacent to the element of C_i that satisfies prepenult is the penultimate element of C_{i-1} if $\beta_{i-1} = 1$ or the bottom element of C_{i-1} if $\beta_{i-1} = 2$. But neither of these elements is
560 in domain . Finally, if $\beta_{i-1} = 3$, then the only elements of C_{i-1} that are adjacent to the element of C_i that satisfies prepenult are the penultimate and bottom elements of C_{i-1} ; but again, neither of these elements is in domain .

5. $H\text{-edge}(x, y)$: This predicate is true of x, y in H_n if both $\text{domain}(x)$
565 and $\text{domain}(y)$ hold, x and y are in the same row and adjacent columns of H_n and either (i) $x \in C_i$ and $y \in C_{i+1}$ for some i ; or (ii) $y \in C_i$ and $x \in C_{i+1}$ with $\beta_i = \beta_{i+1} \notin \{2, 3\}$.

$$\begin{aligned}
\text{H-edge}(x, y) &:= \text{domain}(x) \wedge \text{domain}(y) \wedge \bigwedge_{i=0}^{i=3} \text{Colour}_i(x) \rightarrow \gamma_i(x, y) \\
\gamma_3(x, y) &:= \text{rhscolumn}[3; \{0, 1, 2, 3\}](x, y) \wedge E(x, y) \wedge \\
&\quad \forall z(\text{rhscolumn}[3, \{0, 1, 2, 3\}](x, z) \wedge \\
&\quad \text{lessthan}(y, z, x) \wedge z \neq y) \rightarrow \neg E(x, z) \\
\text{lessthan}(y, z, x) &:= \forall v(\text{samecolumn}(x, v) \wedge E(z, v)) \rightarrow E(y, v) \\
\gamma_2(x, y) &:= \text{rhscolumn}[2; \{0, 1, 2, 3\}](x, y) \wedge E(x, y) \wedge \\
&\quad \forall z(\text{rhscolumn}[2, \{0, 1, 2, 3\}](x, z) \wedge \\
&\quad \text{lessthan}(y, z, x) \wedge z \neq y) \rightarrow \neg E(x, z) \\
\gamma_1(x, y) &:= (\text{Colour}_1(y) \wedge \\
&\quad (\exists z(\text{samecolumn}(y, z) \wedge E(x, z))) \wedge \neg E(x, y)) \\
&\quad \vee (\text{rhscolumn}[1; \{0, 2, 3\}](x, y) \wedge \neg E(x, y)) \\
\gamma_0(x, y) &:= (\text{Colour}_0(y) \wedge E(x, y)) \vee \\
&\quad (\text{rhscolumn}[0; \{1, 2, 3\}](x, y) \wedge E(x, y))
\end{aligned}$$

Suppose $x \in C_i$ and $y \in C_j$. In all cases in the definition above except when
570 $\beta_i = \beta_j \in \{0, 1\}$, it is the case that $\text{rhscolumn}[\cdot](x, y)$ is true, which means
 $j = i+1$. In the case when $\beta_i = \beta_j = 0$, we see that x and y are required to be
adjacent, and when $\beta_i = \beta_j = 1$, x and y are required to be non-adjacent
with the additional condition that there is some element z in the same
column as y that is adjacent to x – both of these cases can happen only
575 when x and y are in adjacent columns and in the same row. We therefore
are left with arguing that when $j = i+1$, then x and y satisfy $\text{H-edge}(x, y)$
if, and only if, they are in the same row.

If $x \in C_i$ and $\beta_i = \{0, 1\}$, the element y of C_{i+1} that is in the same row
as x is easily distinguished. If $\beta_i = 0$, y is the only element of C_{i+1} that is
580 adjacent to x and if $\beta_i = 1$ it is the only element of C_{i+1} not adjacent to x .
When $\beta_i = 3$, we see that for elements z and y of C_{i+1} appearing in say
the rows j and j' , we have $j \leq j'$ if, and only if, every element of C_i (the
column of x) that is adjacent to z is also adjacent to y . This is expressed

by the predicate $\text{lessthan}(y, z, x)$. With this linear order on C_{i+1} defined,
 585 we see that an element y of C_{i+1} is in the same row as x if, and only if, x
 and y are adjacent, and no element of C_{i+1} that is less than y is adjacent
 to x . Analogous arguments can be given for the final case of $\beta_i = 2$.

6. **V-edge** (x, y) : This predicate is true of x, y in H_n if $\text{domain}(x)$ and $\text{domain}(y)$
 both hold, and for some i, j , both x and y appear in the column C_j such
 590 that $\beta_i \neq 0$, and x appears in row i and y in row $i + 1$. In the following
 definition, **TCH-edge** (x, y) denotes that the pair (x, y) is in the reflexive
 and transitive closure of **H-edge**. The reflexive and transitive closure of
 any binary relation is easily defined in MSO.

$$\begin{aligned} \text{V-edge}(x, y) &:= \neg \text{Colour}_0(x) \wedge \text{samecolumn}(x, y) \wedge \\ &\quad \exists u \exists v (\text{prepenultedge}(u, v) \wedge \\ &\quad \quad (\text{TCH-edge}(u, x) \wedge \text{TCH-edge}(v, y))) \\ \text{prepenultedge}(u, v) &:= \neg \text{Colour}_0(u) \wedge \text{samecolumn}(u, v) \wedge \text{prepenult}(v) \wedge \\ &\quad \exists z (\text{prepenult}(z) \wedge \text{H-edge}(z, u)) \end{aligned}$$

The formula **prepenultedge** (u, v) defines those pairs (u, v) in the domain
 595 for which for some i, j , the vertices u, v belong to C_i , and appear resp.
 in the rows j and $j + 1$, with the bottom element of C_i appearing in
 row $j + 3$. To see why this definition is correct, note that **prepenult** (v) is
 true precisely when v appears in row $j + 1$ in C_i (if the bottom element
 of C_i appears in row $j + 3$). To identify u in row j of C_i , we exploit
 600 crucially the special way in which the columns were chosen in G_n : if
 $\beta_i \in \{1, 2, 3\}$ then $|C_i| = |C_{i-1}| + 1$. This ensures that **H-edge** (z, u) holds for
 the element $z \in C_{i-1}$ for which **prepenult** (z) holds. Given the definition of
prepenultedge (u, v) we see that every pair (x, y) of elements in the domain
 that are in the same column and in consecutive rows, is just a “horizontal
 605 translate” of a pair (u, v) satisfying **prepenultedge** (u, v) . That is, x and y
 are reachable from u and v respectively by **H-edge**-paths.

We are now ready to define the MSO interpretation Ψ . Define an “upper
 triangular” $r \times r$ grid as the graph U_r whose vertex set is $\{u_{i,j} \mid 1 \leq j \leq r, i \leq j\}$

and whose edge set is $\{\{u_{i,j}, u_{i,j+1}\} \mid 1 \leq j < r, i \leq j\} \cup \{\{u_{i,j}, u_{i+1,j}\} \mid 1 <$
610 $j \leq r, i < j\}$. A *uniform subdivision* of U_r is the graph obtained by choosing a
subset $S \subseteq \{1, \dots, r-1\}$ and for each $j \in S$ and each $i \leq j$, replacing the edge
 $\{u_{i,j}, u_{i,j+1}\}$ with a path on k_j vertices for some $k_j \geq 2$. It is easy to show that
there exists a parameterless MSO interpretation Γ from graphs to graphs such
that if Z is a uniform subdivision of U_r , then $\Gamma(Z)$ is U_r . Observe that U_{2r}
615 contains the $r \times r$ grid as an induced subgraph.

We now define Ψ as the composition given by $\Psi = \Gamma \circ \Delta$ where $\Delta = (\Delta_V(x),$
 $\Delta_E(x, y))$ is as below. The formulae below contain the predicates $\text{Colour}_0, \text{Colour}_1,$
 $\text{Colour}_2, \text{Colour}_3, \text{top}, \text{topsucc}, \text{bottom}, \text{penult}, \text{prepenult}, \text{first}$ and last which con-
stitute the parameters of Ψ .

$$\begin{aligned}\Delta_V(x) &:= \text{domain}(x) \\ \Delta_E(x, y) &:= \text{H-edge}(x, y) \vee \text{H-edge}(y, x) \vee \text{V-edge}(x, y) \vee \text{V-edge}(y, x)\end{aligned}$$

620 We observe that for the graph H_n defined above, $\Delta(H_n)$ is indeed isomorphic
to a uniform subdivision of U_{2n} . Then $\Psi(H_n)$ is isomorphic to U_{2n} and hence
contains the $n \times n$ grid as an induced subgraph.

Proof of Theorem 4.1. Given $\alpha \in \{0, 1, 2, 3\}^\omega$ containing infinitely many non-
zero letters, consider the MSO interpretation Ψ as described above. The hered-
625 itary closure of $\Psi(\mathcal{S}_\alpha)$ contains the class of all square grids. Taking $\mathcal{C}$ in
Proposition 2.1 to be the class of all square grids, $\mathcal{D}$ to be $\mathcal{S}_\alpha$ and Ξ to be Ψ ,
we are indeed done. $\square$

4.2. Composing Interpretations

We now consider the classes of graphs shown to be minimal HUCW in [10, 13],
630 and prove that these interpret grids using Theorem 4.1 above. Specifically, we
show that for each class $\mathcal{C}$ among them, there is some $\alpha \in \{0, 1, 2, 3\}^\omega$ and an
MSO interpretation Ξ such that the hereditary closure of $\Xi(\mathcal{C})$ contains $\mathcal{S}_\alpha$.
Thus $\mathcal{C}$ interprets grids by Proposition 2.1.

Theorem 4.2. *The following minimal HUCW classes of graphs interpret grids:*

635

1. *Bichain graphs*

2. *Split permutation graphs*

3. *Bipartite permutation graphs*

4. *Unit interval graphs*

Remark 4.3. Note that Theorem 4.2(4) follows from the results of Courcelle
 640 in [15]. It is shown in [15] that Seese's conjecture holds for the class of interval
 graphs. More specifically, it can be inferred from the results in [15] that any
 unbounded clique-width subclass of interval graphs admits MSO interpretability
 of grids. It follows, in particular, that this is true of the unit interval graphs.
 We therefore show parts (1)–(3) of Theorem 4.2 to complete its proof.

645 **Bichain graphs.** We need some terminology to talk about these graphs.
 Given a graph G , a sequence $v_1, \dots, v_k$ of vertices of G is said to be a *chain*
 if $N(v_i) \subseteq N(v_j)$ whenever $i \leq j$, where $N(v) := \{u \mid E(u, v)\}$ denotes the
 neighbourhood of v . A bipartite graph $(A \cup B, E)$ is called a *k-chain graph* if
 each of the two parts A and B can be further partitioned into at most k chains.
 650 A *bichain graph* is a 2-chain graph.

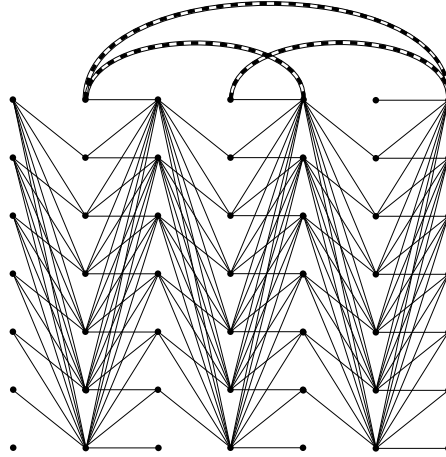


Figure 2: The bichain graph Z_7 . A dashed line between any 2 columns denotes that the
 subgraph induced by the vertices of the two columns is a complete bipartite graph.

We now describe the bichain graph Z_n as defined in [13]. The graph has vertex set $\{z_{i,j} \mid 1 \leq i \leq n, 1 \leq j \leq n\}$ (which can thus be seen as an $n \times n$ grid of points), and $\{z_{i,j}, z_{i',j'}\}$ is an edge if, and only if, one of the following holds: (i) j is odd, $j' = j + 1$ and $i < i'$; (ii) j is even, $j' = j + 1$ and $i' \leq i$; or (iii) j is even, j' is odd and $j' \geq j + 3$. The graph Z_7 is depicted in Figure 2. The graph Z_n is *n-universal* in that all bichain graphs on at most n vertices are induced subgraphs of Z_n . It follows that the class **Bichain** of all bichain graphs is exactly the hereditary closure of the class $\{Z_n \mid n \geq 1\}$.

Again, the grid structure is implicit in the graph Z_n . What we show is that when Z_n is expanded with unary relations for the bottom row $\{z_{n,j} \mid 1 \leq j \leq n\}$ and the last column $\{z_{i,n} \mid 1 \leq i \leq n\}$, we can construct an FO interpretation to a class that contains the class $\mathcal{S}_\alpha$ for $\alpha = (23)^\omega$ in its hereditary closure. We describe below the construction of this interpretation that we denote Ψ .

Let H_n denote the expansion of Z_n with unary predicates **bottom** and **last** that are respectively interpreted as the bottom row of Z_n (namely the set $\{z_{n,j} \mid 1 \leq j \leq n\}$) and the last column of Z_n (so the set $\{z_{i,n} \mid 1 \leq i \leq n\}$). Towards the construction of Ψ , we need the auxiliary predicates **samecolumn**(x, y) and **adjcolumn**(x, y). The first of these is true of x, y in H_n if, and only if, x and y appear in the same column of H_n and neither is the bottom element of that column. The second of these is true if x and y are in adjacent columns in H_n and neither is the bottom element of its column. We assume below that $n \geq 3$.

$$\begin{aligned} \text{samecolumn}(x, y) &:= \neg(\text{bottom}(x) \vee \text{bottom}(y)) \wedge \\ &\quad \forall z(\text{bottom}(z) \rightarrow (E(x, z) \leftrightarrow E(y, z))) \\ \text{adjcolumn}(x, y) &:= \neg(\text{bottom}(x) \vee \text{bottom}(y)) \wedge \\ &\quad \exists u \exists v(\text{samecolumn}(u, x) \wedge \text{samecolumn}(v, y) \wedge E(u, v)) \wedge \\ &\quad \exists u \exists v(\text{samecolumn}(u, x) \wedge \text{samecolumn}(v, y) \wedge \neg E(u, v)) \end{aligned}$$

We briefly reason the correctness of the above predicate definitions. For **samecolumn**(x, y), it is clear that this formula is true for any x and y in the same column of H_n as long as they are not bottom elements. To see that no other pair satisfies the formula, let $x = z_{i,j}$ and $y = z_{i',j'}$ with $j < j'$. We argue

by cases. If j' is odd, then y is adjacent to the bottom element u of column $j' + 1$. Moreover, since $j' + 1$ is then even, u is not adjacent to any $z_{i,j}$ with $j < j'$. On the other hand, if j' is even, then we consider whether j is odd or even. If j is odd, x is adjacent to the bottom element of column $j + 1$ and y is not while if j is even, x is adjacent to the bottom element of column $j' + 1$ and y is not.

For $\text{adjcolumn}(x, y)$, if x and y are in adjacent columns and not bottom elements of their respective columns, then let x' and x'' be the elements in the column of x in resp. the top row and the row just before the bottom in H_n , and let y'' be the element corresponding to x'' in the column of y . Since $n \geq 3$, we have that x', x'', y'' are all distinct. We now see that if the column of x is odd, then $E(x'', y'')$ is false while $E(x', y'')$ is true in H_n ; otherwise, $E(x'', y'')$ is true while $E(x', y'')$ is false in H_n . Then $\text{adjcolumn}(x, y)$ is true in H_n . To see that no other pairs (x, y) other than those just considered satisfy $\text{adjcolumn}(x, y)$, let $x = z_{i,j}$ and $y = z_{i',j'}$ with $j < j' + 1$. We again argue by cases. If j is odd, then since the only column $k \geq j$ for which some vertex of column j is adjacent to some vertex of column k , is the column $k = j + 1$, it follows that $\text{adjcolumn}(x, y)$ is false. If j is even, then if j' is even, then no vertex of column j is adjacent to any vertex of column j' , and if j' is odd, then every vertex of column j is adjacent to every vertex of column j' . In either case, $\text{adjcolumn}(x, y)$ is false.

Consider now the interpretation $\Psi = (\Psi_V, \Psi_E)$ defined as:

$$\begin{aligned}\Psi_V(x) &:= \neg \text{bottom}(x) \wedge \neg \text{last}(x) \\ \Psi_E(x, y) &:= \text{adjcolumn}(x, y) \wedge E(x, y)\end{aligned}$$

It is easy to see that $\Psi(H_{2n+1})$ is the graph Z_{2n} with the edges connecting non-adjacent columns removed; call this graph Z'_{2n} . Let $z_{i,j}$ be the vertex of Z'_{2n} in row i and column j (in the natural grid in which the vertices of Z'_{2n} are arranged). Consider the subgraph of Z'_{2n} induced by the set V of vertices given by $V = \{z_{i,j} \mid k + 1 \leq i \leq k + n \text{ where } \lfloor \frac{j}{2} \rfloor = k, 1 \leq j \leq n\}$. One verifies that this subgraph is indeed isomorphic to the graph Y_n that is induced by the vertices in the first n rows and first n columns, in the graph $\mathcal{P}_\alpha$ where $\alpha = (23)^\omega$.

Proof of Theorem 4.2(1). Consider the interpretation Ψ as described above (having parameters **bottom** and **last**). The hereditary closure of $\Psi(\text{Bichain})$ contains the hereditary closure of $\{Y_n \mid n \geq 1\}$. The latter class $(\{Y_n \mid n \geq 1\} \downarrow)$ is nothing but $\mathcal{S}_\alpha$ for $\alpha = (23)^\omega$. We are now done by Theorem 4.1 and Proposition 2.1. $\square$

Split permutation graphs. Recall that a *split graph* is a graph G whose vertex set can be partitioned into two sets C and I such that C induces a clique in G and I is an independent set in G . A *permutation graph* is a graph whose vertices represent the domain of a permutation, and each of whose edges determines an inversion in the permutation. Following [13], we use the following characterization of split permutation graphs.

Proposition 4.4 ([13, Prop. 2.3]). *Let G be a split graph given together with a partition of its vertex set into a clique C and an independent set I . Let H be the bipartite graph obtained from G by deleting the edges of C . Then G is a split permutation graph if, and only if, H is a bichain graph.*

Let G be a split permutation graph with (C, I) being a partition of its vertex set into a clique C and an independent set I . Let G^* be the expansion of G with a unary predicate P which is interpreted as the set C . Consider the FO interpretation Ψ which removes from G^* all edges inside P . It is easy to see that $\Psi(G^*)$ is a bichain graph by Proposition 4.4.

Let Ψ be the FO interpretation as described above and **SP** be the class of split permutation graphs. Then $\Psi(\text{SP})$, and hence its hereditary closure, contains the class **Bichain**. We are then done by Theorem 4.2(1) and Proposition 2.1.

Bipartite permutation graphs. These graphs are graphs that are bipartite as well as being permutation graphs. For our purposes, the following characterization is useful. Consider the graph P_n on vertex set $\{v_{i,j} \mid 1 \leq i, j \leq n\}$ where the only edges are between $v_{i,j}$ and $v_{i+1,j'}$ for $j' \leq j$. Then, the class of bipartite permutation graphs is exactly the hereditary closure of the class $\{P_n \mid n \geq 1\}$ [10]. Now, it is easily seen that this class is exactly the class $\mathcal{S}_\alpha$ as described in Section 4.1, for $\alpha = 2^\omega$, and this has been observed in [11]. Thus, Theorem 4.2(3) follows from Theorem 4.1.

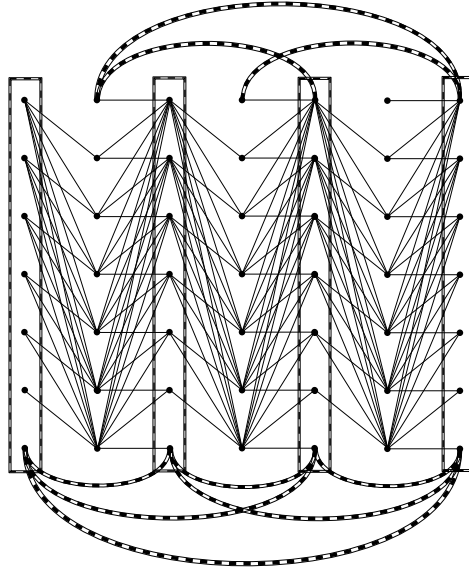


Figure 3: A split permutation graph G . The dashed line surrounding any odd column denotes that the vertices of the column form a clique, and a dashed line between two columns denotes that every vertex of one column is adjacent to every vertex of the other column. One sees that the vertices of the even columns form an independent set in G , while those of the odd columns form a clique. Deleting the edges in this clique gives us the bichain graph Z_7 depicted in Figure 2.

5. Power Graphs

735 In this section, we consider the class of *power graphs* as defined in [14] in the context of well-quasi-ordering and clique-width. Most of the classes that we have seen so far can be shown to not be well-quasi ordered under the induced subgraph relation. In particular, all word-defined classes, unit interval graphs and bipartite permutation graphs can be seen to contain the antichain $\{I_n \mid$
740 $n \geq 1\}$ described after Proposition 3.1. We do not know whether bichain graphs and split permutation graphs are well-quasi ordered, though it has been shown that their expansion with two labels is not a well-quasi ordered class [13]. In contrast, power graphs constitute a class of graphs that is HUCW, that *is* well-quasi ordered [14] and, as we show, is a minimal HUCW class. It was introduced
745 precisely to demonstrate an HUCW class that is well-quasi ordered. Minimality

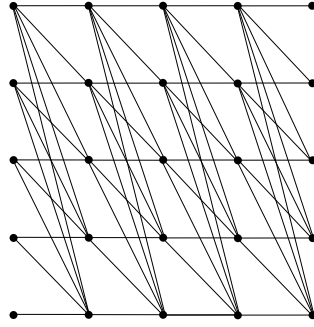


Figure 4: The bipartite permutation graph P_5 . In this illustration the vertex $v_{i,j}$ appears in row j and column i .

follows from arguments contained in [14], but was not observed there. We now define the class of power graphs. We show that they are minimal and then in the remainder of the section show that they admit interpretability of grids.

For $n \geq 1$, we define the graph D_n as follows. The vertex set of D_n is
750 $[n] = \{1, \dots, n\}$. For each $i < n$, there is an edge between i and $i + 1$ —we call these *path edges*. Furthermore, there is an edge between i and j if the largest power of 2 that divides i is the same as the largest power of 2 that divides j —we call these *clique edges*. To understand this terminology, note that we can see D_n as consisting of a simple path of length n , along with, for each k such
755 that $2^k \leq n$, a clique on all vertices $j = 2^k \cdot (2r + 1)$ for some $r \geq 0$ —we call this the *power clique* corresponding to k . In particular, taking $k = 0$, there is a clique formed by all the odd elements, which we call the *odd clique*. An example graph is illustrated in Figure 5. Observe that the path edges, which are the only edges with endpoints in different power cliques always have one end point in the odd
760 clique, and one outside it. The class of power graphs, denoted **Power-graphs**, is now defined as the hereditary closure of the class $\{D_n \mid n \geq 1\}$.

5.1. Minimality of Power Graphs

Proposition 5.1. *The class **Power-graphs** is a minimal hereditary class of unbounded clique-width.*

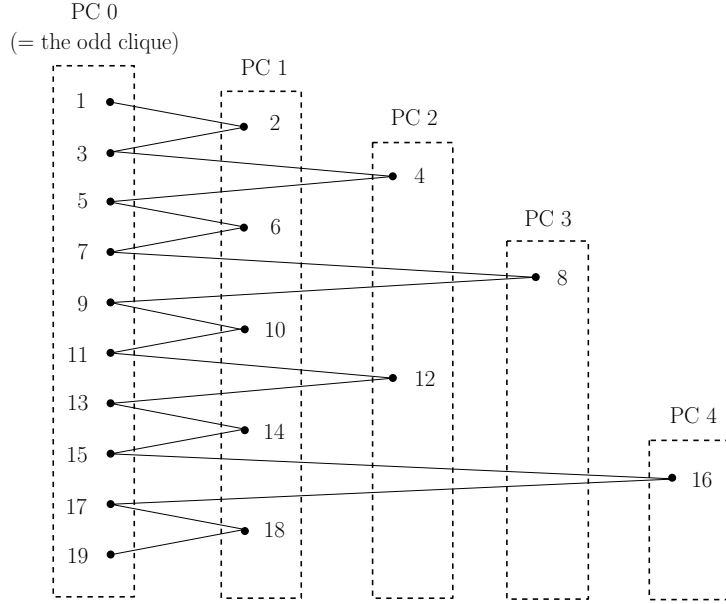


Figure 5: The power graph D_{19} with “PC i ” denoting the power clique corresponding to i .

765 That **Power-graphs** is a hereditary class of unbounded clique-width has already been shown in [14]. Thus, we only need to show that no proper subclass has this property.

Given a graph $G \in \text{Power-graphs}$ which is a subgraph of D_n , define an *interval* in G to be a set $S \subseteq [n]$ of vertices of G such that if $i, j \in S$ with $i < j$ and k is a vertex of G with $i < k < j$ then $k \in S$. We call a subgraph of G induced by an
770 interval a *factor* of G . We now recall the following two results proved in [14].

Lemma 5.2 (Lemma 11, [14]). *Let G be a graph in **Power-graphs**. Then there exists an integer $t = t(G)$ such that for any $n \geq t$, every factor of D_n of length at least t contains G as an induced subgraph.*

775 **Theorem 5.3** (Theorem 2, [14]). *Let G be a graph in **Power-graphs** such that the length of the longest factor in G is t . Then the clique-width of G is at most $2(\log t + 4)$.*

Proof of Proposition 5.1. Consider a proper hereditary subclass $\mathcal{S}$ of **Power-graphs**; then $\mathcal{S}$ excludes a graph $G \in \text{Power-graphs}$. Let $t = t(G)$ be

780 as given by Lemma 5.2. Let $\mathcal{S} = \mathcal{S}_1 \cup \mathcal{S}_2$ where $\mathcal{S}_1 = \mathcal{S} \cap \{D_n \mid n < t\} \downarrow$
and $\mathcal{S}_2 = \mathcal{S} \cap \{D_n \mid n \geq t\} \downarrow$. Observe that $\mathcal{S}_1$ has finitely many graphs up to
isomorphism.

We show that for each $X \in \mathcal{S}_2$, every factor of X has length $< t$. For other-
wise X has a factor Y of length $\geq t$ and there is $p \geq 1$ such that $X \subseteq D_p$ and so Y
785 is also a factor of D_p . Hence by Lemma 5.2, we have G is an induced subgraph
of Y , whereby it is also an induced subgraph of X . Since $\mathcal{S}$ is hereditary, $G \in \mathcal{S}$
which is a contradiction.

By Theorem 5.3, every $X \in \mathcal{S}_2$ has clique-width $\leq k = 2(\log t + 4)$. Then $\mathcal{S}_2$
has bounded clique-width, and hence so does $\mathcal{S}$ since $\mathcal{S}_1$ is finite. $\square$

790 5.2. Interpreting grids in Power Graphs

We now establish the main result of this section, showing that power graphs
do not provide a counter-example to Seese's conjecture.

Theorem 5.4. *There exists an MSO interpretation Θ such that $\Theta(\text{Power-graphs})$
contains all square grids.*

795 We show Theorem 5.4 by showing that there exists an MSO interpretation Φ
such that the hereditary closure of $\Phi(\text{Power-graphs})$ contains all bipartite permu-
tation graphs. We are then done by Theorem 4.2 and Proposition 2.1. Indeed,
it suffices to show that we can interpret grids in a subset of **Power-graphs**, and
we do this for the set $\{D_n \mid n \in \mathbb{N} \text{ even and } n > 9\}$. We first describe the overall
800 ideas involved in the construction of Φ , and provide the details subsequently.

We first show that there exists an FO formula $\text{odd}(x)$ such that if x is a
number in D_n with $n \geq 9$, then $\text{odd}(x)$ is true if, and only if, x is an odd
number. With this formula at hand, we can distinguish path edges from clique
edges. Indeed, an edge is a path edge if, and only if, it has exactly one end
805 point that is odd. In D_n , the path edges form a simple path of length $n - 1$ and,
if n is even, then only one of the two end points satisfies $\text{odd}(x)$. This allows
us to give this simple path an orientation: for each path edge $(x, x + 1)$ we can
identify the direction $x \rightarrow x + 1$. The transitive closure of this relation (which is
definable in MSO), gives us a definition of the natural linear order on D_n .

810 Once we have defined a linear order $\leq$ on D_n , this induces a linear order on
the power cliques: namely, a clique C is below C' if the $\leq$ -minimal element of C
is less than the $\leq$ -minimal element of C' . Indeed, we can also define a successor
relation on cliques from this. From these, we define a relation that relates a
pair x and y precisely if y occurs after x in the linear order $\leq$ and occurs in the
815 power clique that is successor to the power clique containing x . It is easy to
see that the graph induced by this relation contains arbitrarily large bipartite
permutation graphs P_k as defined on page 29.

We now give the details of the construction described above. In addition to
 $\text{odd}(x)$, we need a number of auxiliary predicates along the way.

820 1. We first define the FO formula $\text{odd}(x)$.

$$\text{odd}(x) \quad := \quad \exists y \exists z \exists w \left(\text{"}x, y, z, w \text{ form a 4-clique except for the } z - w \text{ edge"} \right)$$

It is easy to see that for $n \geq 9$, all odd numbers in D_n satisfy $\text{odd}(x)$. If x
is odd with $x < n - 3$, this is witnessed by $y = x + 2$, $w = x + 4$ and $z = x + 1$,
otherwise by $y = x - 2$, $w = x - 4$ and $z = x - 1$.

To show that the even numbers of D_n do not satisfy $\text{odd}(x)$, first observe
825 that in any power clique other than the odd clique, since the numbers in
the clique are of the form $2^k \cdot (2r + 1)$ for fixed k , the difference between
any two numbers in the clique is at least 2^{k+1} , which is at least 4 since
 $k \geq 1$. Suppose now that x is an even number in D_n and x, y, z form a
3-clique. We argue that any w that is adjacent to both x and y must also
830 be adjacent to z showing that $\text{odd}(x)$ is not satisfied. Consider the two
cases:

- The edge between x and y is a clique edge. Then $|x - y| \geq 4$. If z is
in a different power clique, then $|x - z| = 1$ and $|z - y| = 1$, whereby
 $|x - y| \leq 2$ – a contradiction. Thus z is in the same power clique as x
835 and y . By the same argument, w is in the same power clique as x and y ,
so there is a clique edge $z - w$.

• The edge between x and y is a path edge and so $|x - y| = 1$. Then the edges from z to x and y cannot both be path edges, as you cannot have a triangle of such edges. So, one of them is a clique edge. If z is in the same power clique as x , then $|x - z| \geq 4$ and $|y - z| = 1$, which is a contradiction, so z must be in the same power clique as y . By the same argument, w is in the same clique as y , so there is a clique edge $z - w$.

Remark 5.5. The formula $\text{odd}(x)$ is central to our construction below and we assume henceforth that $n \geq 9$.

2. $\text{clique}(x, y)$ and $\text{pathedge}(x, y)$: The formula $\text{clique}(x, y)$ is true of the pair (x, y) in D_n if, and only if, x and y are in the same power clique. The formula $\text{pathedge}(x, y)$ is true if, and only if, $|x - y| = 1$.

$$\begin{aligned}\text{pathedge}(x, y) &:= E(x, y) \wedge (\text{odd}(x) \oplus \text{odd}(y)) \\ \text{clique}(x, y) &:= E(x, y) \wedge \neg \text{pathedge}(x, y)\end{aligned}$$

3. $\text{path}(P, x, y)$: This predicate is true of all triples (P, x, y) for an MSO variable P and $x, y \in D_n$ if P is the (unique) path between x and y , whose edges are all path edges. Below $\exists!w$ denotes “there is a unique w such that...”.

$$\begin{aligned}\text{path}(P, x, y) &:= (P(x) \wedge P(y) \wedge \\ &\quad \exists!w(P(w) \wedge \text{pathedge}(x, w)) \wedge \\ &\quad \exists!w(P(w) \wedge \text{pathedge}(y, w)) \wedge \\ &\quad \forall w((P(w) \wedge w \neq x \wedge w \neq y) \rightarrow \\ &\quad \quad \exists u \exists v(P(u) \wedge P(v) \wedge \text{pathedge}(u, w) \wedge \\ &\quad \quad \text{pathedge}(v, w) \wedge u \neq v))\end{aligned}$$

4. $\text{between}(x, y, z)$: This predicate is true of all triples (x, y, z) in D_n such that y appears somewhere along the (unique) path between x and z (y could be one of x or z).

$$\text{between}(x, y, z) := \exists P(\text{path}(P, x, z) \wedge P(y))$$

We now make a few observations about D_n . Note that since the path edges are definable, and they form a simple path from 1 to n , the only possible automorphisms are the trivial one and the map that reverses the order, in particular mapping n to 1. Moreover, since the odd numbers are definable, for the order
860 reversing map to be an automorphism, n must be odd. We can say more: a more careful analysis shows that the order reversal preserves all power cliques if, and only if, $n = 2^k - 1$ for some k . However, for our purposes it suffices to note that whenever n is even D_n has no non-trivial automorphisms. The predicates we define next are for *even* $n > 9$.

- 865 5. **one**(x): This predicate is satisfied by x in D_n if, and only if, $x = 1$. It defines the unique (when n is even) odd element that has only one path edge incident on it.

$$\text{one}(x) \quad := \quad \text{odd}(x) \wedge \neg \exists z_1 \exists z_2 (\text{pathedge}(x, z_1) \wedge \text{pathedge}(x, z_2) \wedge z_1 \neq z_2)$$

This now allows us to orient the path edges to obtain the natural successor relation on D_n .

- 870 6. **succ**(x, y): This predicate is satisfied by x, y in D_n if, and only if, $y = x + 1$.

$$\text{succ}(x, y) \quad := \quad \text{pathedge}(x, y) \wedge \exists z (\text{one}(z) \wedge \text{between}(z, x, y))$$

As usual, we can then define in MSO a formula **linord**(x, y) which defines the reflexive and transitive closure of **succ**.

7. **cliquemin**(x): This predicate is true of x in D_n if, and only if, x is the minimum element of its power clique (i.e. $x = 2^k$ for some $k \geq 0$).

$$\text{cliquemin}(x) \quad := \quad \forall y (\text{clique}(x, y) \rightarrow \text{linord}(x, y)).$$

875 The linear order defined by **linord** then allows us to linearly order the power cliques.

8. **cliqueord**(x, y): This predicate is true of the pair (x, y) in D_n if, and only if, it is the case that the minimum element in the power clique of x is less

than the minimum element in the power clique of y .

$$\begin{aligned} \text{cliqueord}(x, y) \quad &:= \quad \exists z_1 z_2 (\text{cliquemin}(z_1) \wedge \text{cliquemin}(z_2) \wedge \text{clique}(x, z_1) \\ &\quad \wedge \text{clique}(y, z_2) \wedge \text{linord}(z_1, z_2)) \end{aligned}$$

880 This ordering of the power cliques and the fact that `linord` linearly orders
each clique gives us sufficient structure to define arbitrarily large grids.
To see this concretely, consider the following relation.

9. `cliquemin-succ`(x, y): This predicate is true if x is in the power clique
corresponding to k and y in the power clique corresponding to $k + 1$ for
885 some k .

$$\begin{aligned} \text{cliquemin-succ}(x, y) \quad &:= \quad \neg \text{clique}(x, y) \wedge \text{cliqueord}(x, y) \wedge \\ &\quad \forall z (\text{cliqueord}(x, z) \rightarrow \\ &\quad \quad (\text{clique}(z, y) \vee \text{cliqueord}(y, z))) \end{aligned}$$

10. Consider now the relation `forward`(x, y) defined by

$$\text{forward}(x, y) \quad := \quad \text{cliquemin-succ}(x, y) \wedge \text{linord}(x, y).$$

This relates an element x in the power clique corresponding to k to all
elements of the power clique corresponding to $k + 1$ that are greater than x .

Then, the interpretation $\Phi = (\Phi_V(x), \Phi_E(x, y))$ given by

$$\begin{aligned} \Phi_V(x) \quad &:= \quad \text{True} \\ \Phi_E(x, y) \quad &:= \quad \text{forward}(x, y) \vee \text{forward}(y, x) \end{aligned}$$

890 maps D_n to a graph whose edge relation is the symmetric closure of `forward`.
We claim that the graph $\Phi(D_n)$ contains a large bipartite permutation graph
as an induced subgraph. To see this, choose the largest value k such that the
power clique corresponding to k contains at least k elements in D_n (in other
words $2^k(2k - 1) \leq n$). Consider the subgraph of D_n induced by the set of
895 vertices $\{v_{i,j} \mid 0 \leq i, j \leq k - 1\}$ where $v_{i,j} = j \cdot 2^{k+1} + 2^{i+1}$. Each $v_{i,j}$ is then in the
power clique corresponding to $i + 1$ and it is easily checked that there is an edge
between $v_{i,j}$ and $v_{i',j'}$ in $\Phi(D_n)$ precisely when $i' = i + 1$ and $j \leq j'$.

Proof of Theorem 5.4. As established above, the graph $\Phi(D_n)$ contains an induced subgraph isomorphic to the bipartite permutation graph P_k as long as
900 $2^k(2k - 1) \leq n$, and n is even and at least 9. Then the hereditary closure of $\Phi(\text{Power-graphs})$ contains all bipartite permutation graphs, whereby, by Theorem 4.2 and Proposition 2.1, we are done. $\square$

6. Conclusion

The study of monadic second-order logic on graphs has attracted great attention in recent years. An important aspect of work on this logic is to identify
905 classes of graphs on which MSO is well behaved. Seese’s conjecture is an important focus of this classification effort. In its stronger form it offers a dichotomy: any class of graphs is either interpretable in trees and therefore has bounded clique-width and is well-behaved *or* it interprets arbitrarily large grids and its
910 MSO theory is then undecidable.

We show that Seese’s conjecture could be established by considering two kinds of graph classes: minimal hereditary classes of unbounded clique-width and antichains of unbounded clique-width. Showing that all such classes interpret unbounded grids would suffice. While we do not have a complete taxonomy
915 of such classes, we investigated all the ones known and showed that none of them provides a counter-example to Seese’s conjecture.

We know of only two explicit constructions of antichains of unbounded clique-width: the one presented in this paper and the one due to Korpelainen [21]. Both are explicitly based on grids and easily admit an interpretation of arbitrarily large grids. On the other hand, there is a richer landscape of known
920 minimal HUCW classes and we explore this systematically.

One could weaken the strong conjecture by requiring only that the classes of unbounded clique-width admit MSO *transductions* of grids, rather than interpretations (see [1] for a discussion of transductions). This would still suffice to
925 establish Seese’s conjecture. In all the cases we consider, however, we establish the stronger form, i.e. an interpretation of grids.

It is also worth pointing out that for many of the classes we consider, the original proofs that they have unbounded clique-width require sophisticated bespoke arguments. The interpretation of grids in the classes also provides a
930 uniform method of proving that they have unbounded clique-width.

As a final remark, it is worth noting that there are standard graph operations which allow us to construct new minimal HUCW graph classes from the ones we have. For example, taking the graph complement of all graphs in a class $\mathcal{C}$ yields a class that is also minimal HUCW if $\mathcal{C}$ is. Since this operation is itself an
935 MSO interpretation, the results about interpreting arbitrarily large grids apply to the resulting classes as well.

References

- [1] B. Courcelle, J. Engelfriet, Graph structure and monadic second-order logic: a language-theoretic approach, Vol. 138, Cambridge University Press,
940 2012.
- [2] S. Kreutzer, On the parameterised intractability of monadic second-order logic, in: Computer Science Logic, 23rd international Workshop, CSL 2009, Vol. 5771 of Lecture Notes in Computer Science, Springer, 2009, pp. 348–363. `path(doi : 10.1007/978 - 3 - 642 - 04027 - 6_26)`.
- [3] D. Seese, The structure of the models of decidable monadic theories of graphs, *Annals of pure and applied logic* 53 (2) (1991) 169–195.
945
- [4] B. Courcelle, S. Olariu, Upper bounds to the clique width of graphs, *Discrete Applied Mathematics* 101 (1-3) (2000) 77–114.
- [5] D. G. Corneil, U. Rotics, On the relationship between clique-width and treewidth, *SIAM Journal on Computing* 34 (4) (2005) 825–847.
950
- [6] S.-i. Oum, P. Seymour, Approximating clique-width and branch-width, *Journal of Combinatorial Theory, Series B* 96 (4) (2006) 514–528.

- [7] R. Ganian, P. Hliněný, J. Obdržálek, Clique-width: When hard does not mean impossible, in: T. Schwentick, C. Dürr (Eds.), 28th International Symposium on Theoretical Aspects of Computer Science, STACS 2011, March 10-12, 2011, Dortmund, Germany, Vol. 9 of LIPIcs, Schloss Dagstuhl - Leibniz-Zentrum für Informatik, 2011, pp. 404–415. `path(doi : 10.4230/LIPIcs.STACS.2011.404)`.
URL `https://doi.org/10.4230/LIPIcs.STACS.2011.404`
- [8] J. Nešetřil, P. O. De Mendez, Sparsity: graphs, structures, and algorithms, Vol. 28, Springer Science & Business Media, 2012.
- [9] N. Robertson, P. Seymour, Graph minors. XX. wagner’s conjecture, Journal of Combinatorial Theory, Series B 92 (2) (2004) 325 – 357, special Issue Dedicated to Professor W.T. Tutte.
`path(doi : https://doi.org/10.1016/j.jctb.2004.08.001)`.
URL `http://www.sciencedirect.com/science/article/pii/S0095895604000784`
- [10] V. V. Lozin, Minimal classes of graphs of unbounded clique-width, Annals of Combinatorics 15 (4) (2011) 707–722.
- [11] A. Collins, J. Foniok, N. Korpelainen, V. Lozin, V. Zamaraev, Infinitely many minimal classes of graphs of unbounded clique-width, Discrete Applied Mathematics 248 (2018) 145–152.
- [12] R. Brignall, D. Cocks, Uncountably many minimal hereditary classes of graphs of unbounded clique-width, arXiv preprint arXiv:2104.00412.
- [13] A. Atminas, R. Brignall, V. V. Lozin, J. Stacho, Minimal classes of graphs of unbounded clique-width defined by finitely many forbidden induced subgraphs, arXiv preprint arXiv:1503.01628.
- [14] V. V. Lozin, I. Razgon, V. Zamaraev, Well-quasi-ordering does not imply bounded clique-width, in: International Workshop on Graph-Theoretic Concepts in Computer Science, Springer, 2015, pp. 351–359.

- [15] B. Courcelle, The monadic second-order logic of graphs XV: On a conjecture by D. Seese, *Journal of Applied Logic* 4 (1) (2006) 79 – 114. `path(doi : https://doi.org/10.1016/j.jal.2005.08.004)`.
URL `http://www.sciencedirect.com/science/article/pii/S1570868305000650`
- [16] B. Courcelle, S. il Oum, Vertex-minors, monadic second-order logic, and a conjecture by Seese, *Journal of Combinatorial Theory, Series B* 97 (1) (2007) 91 – 126. `path(doi : https://doi.org/10.1016/j.jctb.2006.04.003)`.
URL `http://www.sciencedirect.com/science/article/pii/S0095895606000463`
- [17] P. Hliněný, D. Seese, Trees, grids, and MSO decidability: From graphs to matroids, *Theoretical computer science* 351 (3) (2006) 372–393.
- [18] W. Hodges, *Model theory*, Cambridge University Press, 1993.
- [19] B. Courcelle, J. Engelfriet, G. Rozenberg, Handle-rewriting hypergraph grammars, *Journal of Computer and System Sciences* 46 (2) (1993) 218 – 270. `path(doi : https://doi.org/10.1016/0022-0000\(93\)90004-G)`.
URL `http://www.sciencedirect.com/science/article/pii/002200009390004G`
- [20] B. Courcelle, From tree-decompositions to clique-width terms, *Discret. Appl. Math.* 248 (2018) 125–144. `path(doi : 10.1016/j.dam.2017.04.040)`.
- [21] N. Korpelainen, A new graph construction of unbounded clique-width, *Electronic Notes in Discrete Mathematics* 56 (2016) 31–36.